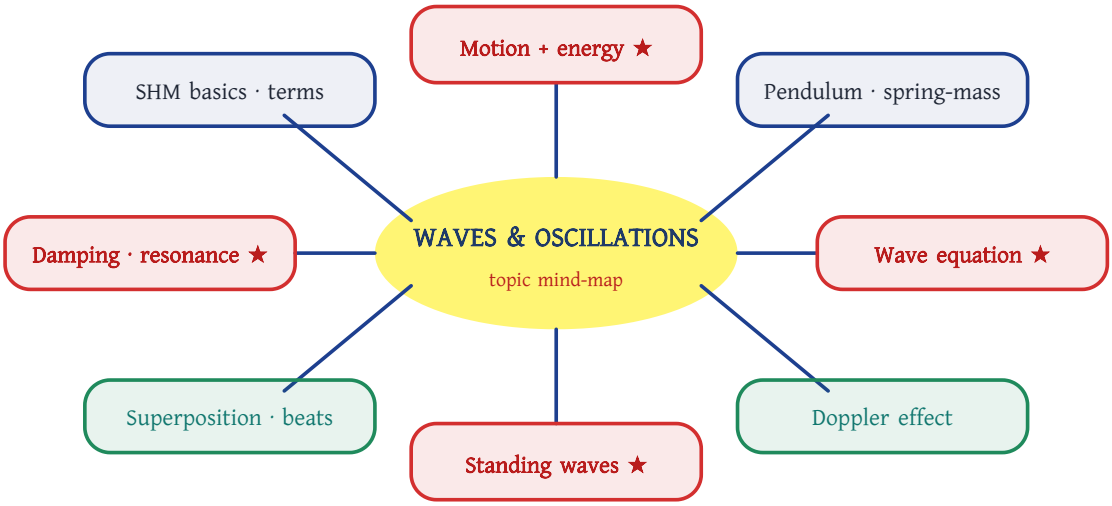
Waves and Oscillations

SHM · Resonance · Wave equation · Standing waves · Doppler — 11 sheets



↑ the whole topic on one page: red = high-yield, green = phenomenon / applied

Syllabus coverage

Basic elements: charge, Coulomb's law … / **Waves and oscillations** / special relativity / electromagnetic waves, photoelectric effect … / de Broglie waves … This notebook covers the syllabus line **"Waves and oscillations"** in full: oscillation and SHM, pendulum & spring systems, damping, forcing and **resonance**, the **wave equation**, **superposition and beats**, **standing waves and harmonics** and the **Doppler effect**. EM-wave properties belong to the Quantum & EM Waves lesson.

High-yield list

① $F = -kx \Rightarrow a = -\omega^2 x$
② $x = A\sin(\omega t + \varphi)$, $E = \frac{1}{2}kA^2$
③ $T = 2\pi\sqrt{(m/k)}$ and $2\pi\sqrt{(L/g)}$
④ $v = f\lambda = \omega/k$, $k = 2\pi/\lambda$
⑤ $f_{beat} = |f_1 - f_2|$
⑥ closed pipe: odd harmonics only
⑦ approach ⇒ higher f

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

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1 Oscillation & Simple Harmonic Motion

Definition: **Oscillation**: a **repetitive to-and-fro motion** of a body about a fixed **mean (equilibrium) position**, repeating after a fixed time. Examples: pendulum, mass on a spring, tuning fork, AC current.

Type	What happens	Example	Type	What happens	Example
Free / natural	no external force	pendulum in air	Undamped	amplitude **constant**	ideal spring
Damped	amplitude **falls**	shock absorber	Forced	external periodic force	radio set
Periodic	repeats after T	all of the above	SHM	$a \propto -x$, **constant A**	spring, pendulum

The five standard quantities:

Amplitude A — max displacement

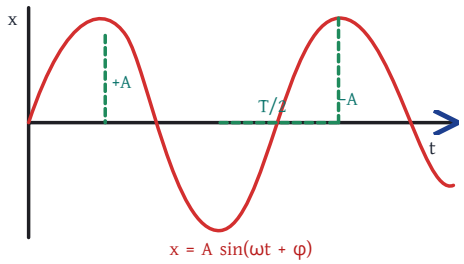
Period T — time of one oscillation

Frequency f — $f = 1/T$, unit **hertz (Hz)** = s^{-1}

Angular freq. ω — $\omega = 2\pi f = 2\pi/T$

Phase φ — start position & direction

one full oscillation = **2π rad = 360°**



Definition: SHM: A periodic motion in which the **restoring force is directly proportional to the displacement and directed toward the mean position**: **F = -kx**. Since F = ma, **$a = -\omega^2 x$** with $\omega^2 = k/m$.

Memory trick: "**M**ean = **M**ax speed, **E**xtreme = **M**ax acceleration": at x = 0 speed is maximum and a = 0; at x = ±A the speed is zero and a is maximum.

Exam trap: Amplitude A is a **displacement**, never the distance travelled: one oscillation covers **4A**. Strictly SHM needs a **constant amplitude**, so a damped pendulum is not SHM.

Worked example ✍: A particle completes **20 oscillations in 8 s**: f = 20/8 = **2.5 Hz**, T = **0.4 s**, ω = 2πf = 5π ≈ **15.7 rad s^{-1}**. With amplitude 4 cm, the distance in 3 oscillations = 3 × 4A = **48 cm**.

IT link: Sampling an oscillating signal needs at least **2f samples per second** (Nyquist), otherwise the frequency is aliased — a 50 Hz mains hum then looks like a low tone.

2 Equations of Motion & Energy in SHM

The three equations (derive them, don't memorise blind):

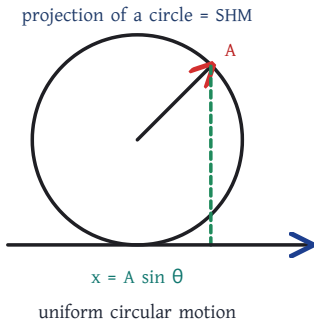
$x = A\sin(\omega t + \varphi)$ $v = dx/dt = A\omega\cos(\omega t + \varphi)$ $a = -\omega^2 x = -A\omega^2\sin(\omega t + \varphi)$

Eliminating t: $\mathbf{v^2 = \omega^2(A^2 - x^2)} \Rightarrow v = \omega\sqrt{(A^2 - x^2)}$.

Position	Velocity	Acceleration	Energy state
x = 0 (mean position)	$v_{max} = A\omega$	a = 0	KE maximum, PE = 0
x = ±A (extreme)	v = 0	$\lvert a\rvert = A\omega^2$	PE maximum, KE = 0

Energy: At displacement x: $\mathbf{KE = \tfrac{1}{2}mv^2 = \tfrac{1}{2}m\omega^2(A^2 - x^2)}$, $\mathbf{PE = \tfrac{1}{2}kx^2 = \tfrac{1}{2}m\omega^2x^2}$. Adding: $\mathbf{E = \tfrac{1}{2}kA^2 = \tfrac{1}{2}m\omega^2A^2}$ — constant wherever the body is.

- Energy is **exchanged** between KE and PE; ideal SHM loses none
- KE and PE each reach their maximum **twice** per complete oscillation
- Energy ∝ $\mathbf{A^2}$: double the amplitude ⇒ **four times** the energy



x	0	A/2	0	-A/2	0	A	0
KE / E	1	¾	1	¾	1	0	1
PE / E	0	¼	0	¼	0	1	0

fraction of the total energy during **one** complete oscillation (x = A sin ωt)

Memory trick: Force is **R**estoring and **P**roportional — hence the minus sign. "**KE** at the **C**entre, **PE** at the **E**nds."

Exam trap: At the extremes the speed is zero but the acceleration is **maximum**. ½kA² is the **total** energy; ½kA and kA² are both wrong.

Worked example ✍️: A = 0.02 m, f = 5 Hz ⇒ ω = 10π ≈ 31.4 rad/s. $v_{max} = A\omega = \mathbf{0.63\ m\ s^{-1}}$; $a_{max} = A\omega^2 = \mathbf{19.7\ m\ s^{-2}}$. At x = A/2, $v = \omega\sqrt{(A^2 - x^2)} = \mathbf{0.54\ m\ s^{-1}}$.

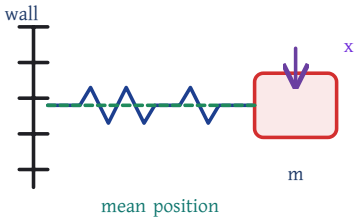
> x = A sin ωt only when x = 0 at t = 0

IT link: A phone accelerometer is a tiny mass–spring oscillator: it measures **a** from x. A vibration tool picks the **fundamental and its harmonics** out of the FFT of the x–t trace.

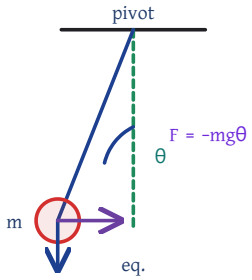
SHM vs circular motion: A body in uniform circular motion is **not** in SHM: its acceleration always points at the centre. Only its *projection* on a diameter is SHM.

3 Two Systems You Must Know

Mass on a spring: Hooke's law F = -kx (k in **N m^{-1}**) ⇒ a = -(k/m)x ⇒ **ω = √(k/m)**, **T = 2π√(m/k)**. Independent of g and of amplitude.



Simple pendulum: F = -mg sinθ ≈ **-mgθ** for small θ (< 10°), and x ≈ Lθ ⇒ a = -(g/L)x ⇒ **ω = √(g/L)**, **T = 2π√(L/g)**. Independent of the bob's mass. A larger amplitude makes T slightly **longer**: T ≈ $T_0(1 + \theta_0^2/16)$, θ_0 in radians.



Feature	Spring–mass	Simple pendulum	Feature	Spring–mass	Simple pendulum
Restoring force	-kx	-mgθ	Increase m	T ↑ by √m	**no change**
Period	**2π√(m/k)**	**2π√(L/g)**	Increase L	—	T ↑ by √L
Depends on	m, k only	L, g only	On the Moon	**no change**	T ↑ by √6 ≈ 2.45

Worked example 1 ✍: m = 0.5 kg, k = 200 N/m ⇒ ω = √(200/0.5) = √400 = **20 rad/s**; T = 2π/20 = **0.314 s**; f ≈ 3.2 Hz.

Worked example 2 ✍: L = 1 m, g = 9.8 ⇒ T = 2π√(1/9.8) = **2.01 s**. Make L four times ⇒ T = √4 × T = **2T**, not 4T.

Memory trick: Spring → watch **k and m**. Pendulum → watch **L and g**: the bob's mass cancels, so an **m** never appears in its T.

Exam trap: L × 4 ⇒ T × 2, **never** × 4; on the Moon T × √6 ≈ 2.45. A spring–mass period has no g, so the Moon changes it not at all.

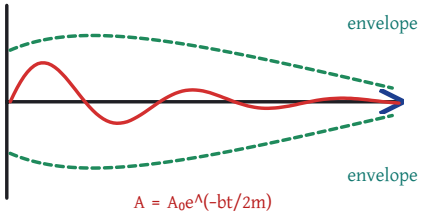
Worked example 3 ✍: Pendulum on the Moon (g = g/6): T = 2π√(L/(g/6)) = **√6 × T ≈ 2.45 T** — the same pendulum beats more slowly. The spring–mass above is **unaffected**.

Worked example 4 ✍: k = 800 N/m with m = 0.2 kg: f = (1/2π)√(800/0.2) = (1/2π)×√4000 = **10.1 Hz**, T ≈ 0.099 s. Four masses in parallel → m = 0.8 kg → f falls by √4 = **2**.

IT link: A shock absorber is a deliberately **damped** mass on a spring; a radio tunes an LC oscillator, $f = 1/2\pi\sqrt{(LC)}$ — the same resonance idea twice.

4 Damped, Forced Oscillation & Resonance

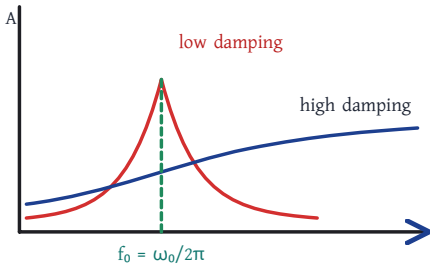
Damped oscillation: A force **opposite to the velocity** (friction, air drag) leaks energy as heat: **$mx'' + bx' + kx = 0$**, amplitude **$A = A_0e^{(-bt/2m)}$**. The **period is unchanged**.



Damping	Where the energy goes
Viscous / air	fluid friction → heat
Coulomb (dry)	constant force, **linear** fall
Electromagnetic	eddy currents → Joule heat

Forced oscillation: A periodic external force $F_0 \cos \omega t$ drives the system: **$mx'' + bx' + kx = F_0 \cos \omega t$**. After the transient dies out the body moves at the **driving frequency**, with amplitude $A(\omega) = (F_0/m) / \sqrt{[(\omega_0^2 - \omega^2)^2 + (b\omega/m)^2]}$.

Resonance: The amplitude is **maximum** when the driving frequency equals the natural frequency: **$\omega = \omega_0 = \sqrt{(k/m)}$**. There the force **lags the displacement by $\pi/2$**, so force is in phase with velocity and energy is pumped in fastest.



Sharpness: $Q = \omega_0 m/b$. Large Q (little damping) ⇒ tall narrow peak; small Q ⇒ broad flat curve. Used in **radio/TV tuning**, the resonance column, MRI and organ pipes. Danger: bridge resonance, marching soldiers, tall buildings in an earthquake.

Memory trick: "Damaged = amplitude **dies**, period **lives**." Resonant peak ⇔ energy pumped in per cycle equals energy lost per cycle.

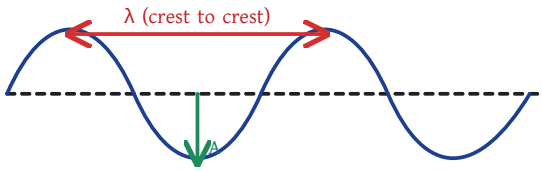
Exam trap: Resonance means **matching frequencies**, not "always loud"; a big amplitude needs low damping. With damping the amplitude peak sits slightly **below** f_0.

Worked example ✍️: $\omega_0 = 20$ rad/s. Driven at 20 rad/s ⇒ resonance. At 40 rad/s, with negligible damping $A \propto 1/(\omega_0^2 - \omega^2)$, so $A \propto 1/(\omega_0^2 - 4\omega_0^2)$ ⇒ the amplitude is **one third** of the resonant value.

Worked example ✍️: Sharpness: $\omega_0 = 20$ rad/s, m = 0.5 kg, b = 2 kg $s^{-1} \Rightarrow Q = \omega_0 m/b = 20 \times 0.5/2 =$ **5**. Halve b and Q doubles: the peak becomes twice as tall and half as wide.

5 Wave Motion & the Wave Equation

Definition: A **wave** is a travelling disturbance that **transfers energy without transferring matter**: the particles of the medium oscillate about their own mean positions while the *pattern* moves on. Sound crosses air at 340 m s^{-1} yet no air molecule travels that far.



TRANSVERSE: particles ⊥ travel



LONGITUDINAL: particles ∥ travel

Type	Examples	Key fact	Type	Examples	Key fact
Mechanical	sound, water, string	needs a **medium**	Transverse	string, light, surface	can be **polarised**
Electromagnetic	light, radio, X-ray	travels in **vacuum**	Longitudinal	sound in air, spring	never polarised

The wave equation and its symbols:

$y(x, t) = A \sin(kx - \omega t + \varphi)$ (A cos is equally valid; the sign only fixes the direction of travel)

A = amplitude (m)

λ = wavelength (m)

$k = 2\pi/\lambda$ = **wave number** (rad m^{-1})

$\omega = 2\pi f$ = angular frequency

$\mathbf{v = f\lambda = \omega/k}$

$\partial^2 y/\partial x^2 = (1/v^2)\partial^2 y/\partial t^2$

$\Delta\varphi = 2\pi\Delta x/\lambda = 2\pi f\Delta t$

string: $v = \sqrt{(F/\mu)}$

Speed of a wave: String $v = \sqrt{(F/\mu)}$ · fluid $v = \sqrt{(B/\rho)}$ · solid rod $v = \sqrt{(Y/\rho)}$. Order $\mathbf{v_{solid} > v_{liquid} > v_{gas}}$; sound in air ≈ **331 m/s** (0 °C), 343 m/s (20 °C); EM waves in vacuum 3×10^8 m/s.

Exam trap: A wave crossing into a new medium keeps the **same frequency** — only v and λ change ($v \propto \lambda$). Energy $\propto \mathbf{A^2}$: double A ⇒ 4× the power. Sound is mechanical: **no sound in vacuum**.

Memory trick: "Go From a Crest to the next Crest" and λ is done — one full oscillation of every point of the medium. $k = 2\pi/\lambda$ is a **wave number** in rad m^{-1}, not a frequency.

Exam trap: A wave is not matter moving: the amplitude of the *medium particles* is not the amplitude of the wave. Wave and particle speeds are unrelated — a long wavelength can travel fast.

Worked example ✍️: f = 500 Hz, λ = 0.68 m ⇒ v = fλ = 500 × 0.68 = **340 m s^{-1}**. In water (v = 1500 m s^{-1}) the frequency stays **500 Hz**, so λ = v/f = **3 m**. · Wi-Fi 2.4 GHz: λ = c/f = $3\times10^8/2.4\times10^9$ = **12.5 cm** — routers use λ/2 or λ/4 antennas.

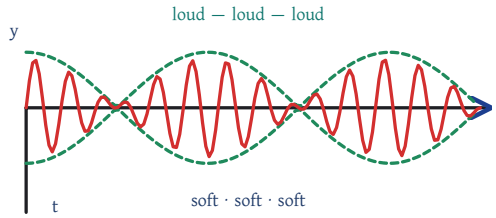
6 Superposition, Interference & Beats

Principle of superposition: When waves meet, the **resultant displacement is the algebraic sum** of the individual displacements. The waves **pass through each other unchanged** and continue on their way — the basis of radio interference, noise cancellation and beam forming.

Case	Path difference Δ	Phase difference	Result
Constructive (in phase)	nλ (n = 0, 1, 2 …)	2nπ	amplitude **maximum**
Destructive (out of phase)	(n + ½)λ	(2n + 1)π	amplitude **minimum** — zero if $A_1 = A_2$

Beats: Two waves of **equal amplitude** and **very nearly equal frequency** superpose: the sum is a fast vibration inside a slow swelling and shrinking envelope, heard as loud → soft → loud.

$y = 2A\cos[\pi(f_1 - f_2)t]\cdot\cos[\pi(f_1 + f_2)t]$ ⇒ **$f_{beat} = |f_1 - f_2|$**, interval between beats = $1 / f_{beat}$.



- Beats are **distinct** only for a small difference (roughly < 20 Hz); $f_1 = f_2$ ⇒ **no** beats, one steady tone
- Uses: detecting an inaudible frequency by beating it with a known one, calibrating a tuning fork, tuning by beat notes

Memory trick: The **beat** frequency is the difference; the **note** you hear is the average: $f_{note} = (f_1 + f_2)/2$, $f_{beat} = |f_1 - f_2|$.

Exam trap: f_{beat} is **not** the average and **not** a signed difference — always take the absolute value. If $|f_1 - f_2| > 20$ Hz you only hear a change in quality.

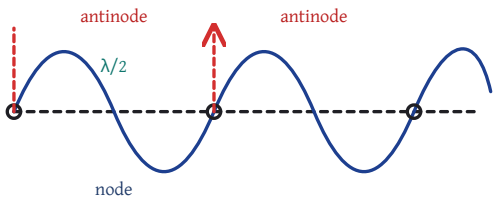
Worked example ✍️: Forks of 440 Hz and 444 Hz: f_{beat} = **4 Hz**, interval 0.25 s, 40 beats in 10 s, average pitch 442 Hz. To test whether an unknown tone is 1000 or 1004 Hz, sound it with a 1000 Hz fork: **4 beats s^{-1}** ⇒ it is the 1004 Hz one.

IT link: Two nearby Wi-Fi access points on the same channel beat against each other (interference → dropouts); superheterodyne radios deliberately mix two frequencies and listen to the *difference* (beat) frequency.

7 Standing Waves, Nodes & Harmonics

Standing wave: Two **identical** waves travelling in **opposite** directions superpose: $A \sin(kx - \omega t) + A \sin(kx + \omega t) = \mathbf{2A \sin kx \cos \omega t}$. The pattern no longer travels — it stands still, so **no energy is transferred along it**.

Point	Displacement	Amplitude
Node	always zero	0
Antinode	oscillates fully	max = 2A
Node → node	—	**$\lambda/2$**
Node → next antinode	—	**$\lambda/4$**



String, both ends fixed: Both ends are nodes: $L = n\lambda/2 \Rightarrow$ **$f_n = nv/2L$**, n = 1, 2, 3 … **all harmonics**. Nodes = n + 1, antinodes = n.

Pipe open at both ends: Both ends are **displacement antinodes**: $L = n\lambda/2 \Rightarrow$ **$f_n = nv/2L$, all harmonics.** Antinodes = n, nodes = n − 1.

Pipe closed at one end: Closed end = node, open end = antinode: $L = (2n - 1)\lambda/4 \Rightarrow$ **$f_n = (2n - 1)v/4L \Rightarrow$ odd harmonics only** (f, 3f, 5f).

System	Fundamental	2nd harmonic	3rd harmonic	Ratio	1st overtone is the
String (both ends fixed)	v/2L	2v/2L	3v/2L	1 : 2 : 3	2nd harmonic
Open pipe (both ends open)	v/2L	2v/2L	3v/2L	1 : 2 : 3	2nd harmonic
Closed pipe (one end shut)	v/4L	absent	3v/4L	1 : 3 : 5	**3rd harmonic**

Memory trick: "**O**pen = all **O**rders allowed (1, 2, 3); **C**losed = **C**ount only the odds (1, 3, 5)." So a closed pipe's first overtone needs **3L**, not 2L.

Exam trap: Node to node is **$\lambda/2$**, not λ. Both ends of a fixed string are **nodes**; both ends of an open pipe are **antinodes**. Standing waves carry no net energy along the medium.

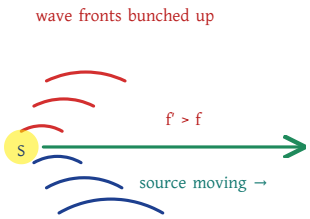
Worked example ✍️: String L = 2 m, μ = 0.5 kg m^{-1}, F = 200 N $\Rightarrow$ $v = \sqrt{(F/\mu)} = \sqrt{400} =$ **20 m s^{-1}**; $f_1 = v/2L =$ **5 Hz**; $f_3 = 3f_1 =$ **15 Hz** ($\lambda_3 = 2L/3 = 1.33$ m, 4 nodes, 3 antinodes). Closed pipe L = 0.25 m, v = 340 $\Rightarrow$ $f_1 = v/4L = 340$ Hz, first overtone **1020 Hz**.

8 The Doppler Effect

Definition: The apparent change in **frequency** (and hence in pitch and observed wavelength) of a wave when there is **relative motion between source and observer**. Christian Doppler, 1842.

Motion	Observed frequency f′	Motion	Observed frequency f′
Source approaching (> f)	$f' = f\,v/(v - v_s)$	Observer approaching (> f)	$f' = f(v + v_o)/v$
Source receding (< f)	$f' = f\,v/(v + v_s)$	Observer receding (< f)	$f' = f(v - v_o)/v$
Both moving	$f' = f(v + v_o)/(v - v_s)$	Light (relativistic)	$f' = f\sqrt{[(1+\beta)/(1-\beta)]}$, $\beta = v/c$

Here **v** is the speed of the wave **in that medium** (340 m s^{-1} for sound in air) and the medium is taken to be at rest. What matters is the compression of the wavelength, so a source and an observer moving together *with* the medium hear nothing unusual.



- **Sound:** a siren's pitch rises then falls as it passes
- **Echo / radar:** the shift is applied twice, out and back — $f_{echo} = f(v + v_o)/(v - v_o)$; police radar guns, Doppler ultrasound
- **Blood flow:** cells moving towards the probe shift the echo up, and the shift measures their speed
- **Light:** galaxy receding ⇒ **redshift** (expanding universe); approaching ⇒ blueshift

Memory trick: "**C**rush in ⇒ **C**rowd": closer wavefronts, **higher** frequency. "**F**lee far ⇒ **F**lat." Whatever the cause — **approach ⇒ high, recede ⇒ low.**

Exam trap: Only the wave speed **in the medium** appears, not the source–observer relative speed. Doppler changes **pitch**, not loudness. For light use $f' = f\sqrt{[(1+\beta)/(1-\beta)]}$.

Worked example ✍️: A 1000 Hz siren at 15 m s^{-1} (v = 340): approaching $f' = 1000 \times 340/325 =$ **1046 Hz**; receding $f' = 340000/355 =$ **958 Hz**. Observer at 20 m s^{-1} towards a still source: $f' = 1000(340 + 20)/340 =$ **1059 Hz**.

Night-Before-the-Exam Sheet

Formula sheet:

$f = 1/T$, $\omega = 2\pi f = 2\pi/T$ (Hz, rad s^{-1})

$x = A\sin(\omega t + \varphi)$

$F = -kx \Rightarrow a = -\omega^2 x$

$v = \omega\sqrt{(A^2 - x^2)}$

$v_{max} = A\omega$, $a_{max} = A\omega^2$

$E = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2A^2$

Spring: $T = 2\pi\sqrt{(m/k)}$

Pendulum: $T = 2\pi\sqrt{(L/g)}$

Damped: $A = A_0e^{\wedge}(-bt/2m)$

Resonance: $\omega = \omega_0$, $Q = \omega_0 m/b$

$v = f\lambda = \omega/k$, $k = 2\pi/\lambda$

$\partial^2y/\partial x^2 = (1/v^2)\partial^2y/\partial t^2$

String: $v = \sqrt{(F/\mu)}$

$\Delta\varphi = 2\pi\Delta x/\lambda = 2\pi f\Delta t$

$f_{beat} = |f_1 - f_2|$

Node → node = $\lambda/2$

String / open pipe: $f_n = nv/2L$

Closed pipe: $f_n = (2n - 1)v/4L$

Doppler source: $f' = fv/(v - v_s)$

Doppler observer: $f' = f(v + v_o)/v$

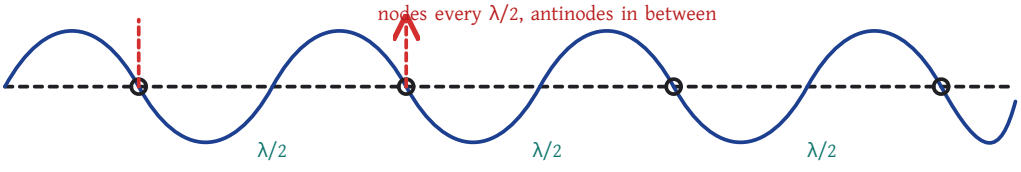
Top 12 one-liners for MCQs

1. SHM: $a \propto -x$, $F = -kx$
2. Unit of frequency: hertz = s^{-1}
3. T of spring $\propto \sqrt{m}$; of pendulum $\propto \sqrt{L}$
4. Pendulum T ignores the bob's mass
5. Total SHM energy = $\frac{1}{2}kA^2$
6. $v_{max} = A\omega$ at the mean position
7. $v = f\lambda = \omega/k$, $k = 2\pi/\lambda$
8. Frequency never changes between media
9. Wave energy $\propto A^2$
10. Beat frequency = difference of the two
11. Closed pipe: odd harmonics only
12. Doppler: approach ⇒ higher frequency

Top traps: Amplitude ≠ distance travelled (that is 4A) · a is maximum at the extremes · $v = \omega\sqrt{(A^2 - x^2)}$, not $\omega(A - x)$ · frequency is fixed when a wave changes medium · beat frequency is the difference, not the average · a closed pipe has no 2nd harmonic · node to node is $\lambda/2$

All tricks together: Mean = max speed, extreme = max acceleration · force is Restoring and Proportional, hence the minus sign · spring: watch k and m; pendulum: watch L and g · damping kills amplitude, never the period · open pipe = every harmonic, closed = odds only · the beat is the difference, the note is the average · crush in ⇒ crowd (higher f)

Sound checkpoints: audible 20 Hz – 20 kHz (below = infrasonic, above = ultrasonic) · speed ≈ 331 m/s at 0 °C and 343 m/s at 20 °C · $v_{solid} > v_{liquid} > v_{gas}$ · intensity $\propto A^2$ · no sound through a vacuum.



the standing-wave ruler: fix a node and the whole pattern follows

✓ Practice MCQs (exam pattern, 10 marks)

1. The SI unit of frequency:
(a) Hertz (b) Joule
(c) Newton (d) Pascal

2. A 440 Hz wave of λ = 2 m travels at:
(a) 220 m/s (b) 880 m/s
(c) 440 m/s (d) 2200 m/s

3. Period of a 2 kg mass on a 200 N/m spring:
(a) 0.314 s (b) 1.57 s
(c) 0.157 s (d) 0.628 s

4. A pendulum made 4× longer has a period:
(a) Half (b) 4 ×
(c) 2 × (d) 1/4

5. Closed pipe resonance ratio:
(a) 1, 2, 3 (b) 1, 3, 5
(c) 1 only (d) 2, 4, 6

6. Beats of 500 Hz and 505 Hz waves:
(a) 5 Hz (b) 1005 Hz
(c) 2.5 Hz (d) 502.5 Hz

7. SHM energy at amplitude A, constant k:
(a) ½kA (b) kA^2
(c) $\frac{1}{2}kA^2$ (d) $\frac{1}{2}mv^2$ only

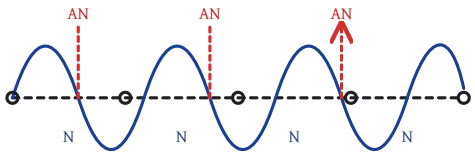
8. Closed pipe length for its 1st overtone:
(a) 3L (b) 2L
(c) 4L (d) L/3

9. Resonance needs:
(a) Max driving force (b) Zero amplitude
(c) Zero frequency (d) Equal frequencies

10. A 500 Hz source approaches at 20 m/s (v = 340 m/s):
(a) 472 Hz (b) 531 Hz
(c) 500 Hz (d) 560 Hz

Answer key → 1-a · 2-b · 3-d · 4-c · 5-b · 6-a · 7-c · 8-a · 9-d · 10-b

Written question: A string of length 2 m and mass per unit length 0.5 kg m^{-1} is fixed at both ends and vibrates under a tension of 200 N. (a) Speed of the waves? (b) Fundamental and third-harmonic frequency? (c) Nodes and antinodes, and wavelength, there? 7 marks



3rd harmonic: 4 nodes, 3 antinodes, λ = 2L/3

1. v = √(F/μ) = √(200/0.5) = √400 = **20 m s^{-1}** 2
2. f_1 = v/2L = 20/4 = **5 Hz** 2
3. $f_3 = 3f_1$ = **15 Hz** 1
4. Nodes = n + 1 = **4** (2 fixed ends + 2 inside), antinodes = n = **3** 1
5. λ_3 = 2L/3 = 4/3 m ≈ **1.33 m** 1

Before the exam ✓

- ✓ All six SHM quantities, with units
- ✓ Derive both 2π√(m/k), 2π√(L/g)
- ✓ Split SHM energy at any x
- ✓ Write y = A sin(kx − ωt + φ)
- ✓ Count nodes and antinodes

Also practise (written):
① First overtone of a closed pipe, L = 0.5 m
② Beats between 256 Hz and 260 Hz
③ Doppler shift for an observer on a train